Srijith Nair

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Research Profile

PhD candidate developing provably convergent, resource-efficient methods for continual and federated learning. First author of FSL-SAGE at ICML 2025, combining convergence analysis with PyTorch/gRPC implementation on heterogeneous, geo-distributed GPU systems; additional first-author work develops memory-efficient continual-learning algorithms with Pareto-optimality guarantees. Broader research includes federated large language model (LLM) fine-tuning, in-context learning theory, and Lean-based evaluation for verifiable code generation.

Research Expertise

Core: continual learning; multi-objective and distributed optimization; federated/split learning; resource-efficient LLM training

Additional: in-context learning theory; inverse problems; formal methods and Lean 4

Education

The Ohio State University — PhD (expected), Electrical and Computer Engineering (GPA: 4.0/4.0) *Aug 2020 – May 2027*
Advisor: Prof. Jia (Kevin) Liu

The Ohio State University — MS, Electrical and Computer Engineering (GPA: 4.0/4.0) *May 2024*
Thesis: *Robust Blind Image Denoising via Instance Normalization*
Advisor: Prof. Philip Schniter

NIT Trichy, India — BTech, Electronics and Communication Engineering *May 2018*

Selected Publications

- Z. Luo, J. Cai, C. Le Denmat, **S. Nair**, F. Nourzad, R. K. Sudha, et al., "Toward WAN-Aware LLM Training Across Heterogeneous, Geo-Distributed Sites," in *Proceedings of the ACM SIGCOMM 2026 Conference*. New York, NY, USA: Association for Computing Machinery, 2026, pp. 2057–2059, ISBN: 9798400724671. [paper].
- **S. Nair**, A. Eryilmaz, and J. Liu, "PMF-CL: Pareto-Minimal Forgetting Continual Learner for Conflicting Tasks," Under review, 2026.
- **S. Nair**, A. Vempaty, J. Liu, and A. Jagamohan, "Beyond Type-checking: Towards Holistic Evaluation of Formal Specification Generation," Under review, 2026.
- H. Liu, R. Wen, **S. Nair**, J. Liu, W. Lou, C. Zhang, et al., "EcoLoRA: Communication-Efficient Federated Fine-Tuning of Large Language Models," in *The 2025 Conference on Empirical Methods in Natural Language Processing*, 2025. [paper].
- **S. Nair**, M. Lin, P. Ju, A. Talebi, E. S. Bentley, and J. Liu, "FSL-SAGE: Accelerating Federated Split Learning via Smashed Activation Gradient Estimation," in *Forty-second International Conference on Machine Learning*, 2025. [paper].

Research Experience

Emergence AI *May 2026 – Aug 2026*
AI Research Intern — New York City, NY

- Developed Lean 4 evaluation metrics for formal specification generation, assessing quality beyond type-checking in verifiable code-generation systems.
- Extended an inductive–deductive approach to verifiable code generation; communicated results through a research preprint and technical blog post.

NSF AI-EDGE Institute, The Ohio State University *May 2024 – present*
Graduate Research Associate — Columbus, OH

- Designed, analyzed, and implemented **FSL-SAGE**, the first provably convergent auxiliary-model method for federated split learning; achieved a 2.5× reduction in communication overhead in the reported evaluation and published the work first-author at ICML 2025.
- Implemented federated LLM fine-tuning in PyTorch/gRPC on a cross-institution, geo-distributed GPU testbed spanning UW–Madison, UT Austin, IIT Bombay, and the University of Michigan; contributed to **EcoLoRA** and a WAN-aware LLM training study.
- Developed **PMF-CL**, a continual-learning meta-algorithm with convergence to Pareto-optimal solutions for conflicting tasks; derived $O(d^2)$ persistent-memory implementations for linear/basis regression and quadratic objectives.
- Proved that ReLU and hard-max multi-head attention layers can implement in-context binary classification; analyzed the roles of positional encoding and context structure.

The Ohio State University *Aug 2020 – Apr 2024*
Graduate Research Associate, Signal Processing — Columbus, OH

- Developed robust blind denoising and unrolled inverse-imaging methods using one-layer spline networks, improving training and memory efficiency while retaining reconstruction quality.
- Derived convergence results for operator-splitting methods with pseudo-contractive operators and designed monotone constraints for learned regularizers; implemented constrained bilevel optimization in JAX.
- Proposed quantile-based normalization for scaling- and shift-equivariant blind image denoising, enabling a model trained at one noise level to generalize across unseen noise levels.

Indian Institute of Science *May 2018 – Jul 2019*
Research Assistant — Bengaluru, India

- Built and deployed SATVAM, a low-cost wireless air-quality monitoring system using 6LoWPAN/RPL and statistical sensor calibration; work appeared at IEEE eScience and in *Atmospheric Measurement Techniques*.
- Designed CONETSI, a decentralized protocol for on-demand state collection in resource-constrained IoT networks; published first-author at IEEE COMSNETS 2020.
- Worked across embedded sensing, low-power networking, and machine-learning-based sensor calibration in a multidisciplinary, cross-institution deployment.

Technical Skills

ML and research: PyTorch, JAX, TensorFlow, NumPy/SciPy, optimization, convergence analysis, experiment design
Systems: distributed training, gRPC, Linux, GPU experimentation, W&B, 6LoWPAN/RPL
Programming: Python, C/C++, Bash, Lean 4, LaTeX

Selected Research Presentations

- **S. Nair**, M. Lin, P. Ju, A. Talebi, E. Bentley, and J. Liu, "FSL-SAGE: Accelerating federated split learning via smashed activation gradient estimation," in *International Conference on Machine Learning*, 2025. [poster].
- **S. Nair** and P. Schniter, "An improved spline-based learned convex regularizer for image recovery," in *Asilomar Conference on Signals, Systems, and Computers*, 2023. [poster].
- **S. Nair** and P. Schniter, "Deep regularization via bi-level optimization with adversarial regularization," in *BASP Frontiers*, 2023. [poster].
- **S. Nair** and P. Schniter, "Deep regularization via bi-level optimization," in *Asilomar Conference on Signals, Systems, and Computers*, 2022. [poster].

Additional Publications

- R. Sahu, A. Nagal, K. K. Dixit, H. Unnibhavi, S. Mantravadi, **S. Nair**, et al., "Robust statistical calibration and characterization of portable low-cost air quality monitoring sensors to quantify real-time O_3 and NO_2 concentrations in diverse environments," *Atmospheric Measurement Techniques*, vol. 14, no. 1, pp. 37–52, 2021. [paper].
- **S. Nair**, A. Sahu, S. Mantravadi, S. Anand, and M. Hegde, "CONETSI: On-Demand Distributed Network State Information Collection using Opportunistic Exploration for Resource Constrained Networks," in *2020 International Conference on COMmunication Systems & NETworkS (COMSNETS)*, 2020, pp. 491–498. [paper].
- Y. Simmhan, **S. Nair**, S. Monga, R. Sahu, K. Dixit, R. Sutaria, et al., "SATVAM: Toward an IoT Cyber-Infrastructure for Low-Cost Urban Air Quality Monitoring," in *2019 15th International Conference on eScience (eScience)*, 2019, pp. 57–66. [paper].

Recognition and Service

Award: Dr. Burn Lin Graduate Travel Grant, ICML 2025
Hackathon: First Place, Make OHI/O 2022 — refreshable Braille display
Peer review: NeurIPS, ICML, ICLR, IEEE/ACM Transactions on Networking, IEEE TMC, IEEE TPAMI

Teaching and Mentoring

Graduate Teaching Associate for *Introduction to Machine Learning* and *Discrete Signals and Systems* at The Ohio State University (2023–2024); led labs and office hours and graded assignments, lab reports, and exams.